YAKEEN

NEET 2.0 2027

Physics | Lecture - 4

Motion in a Plane

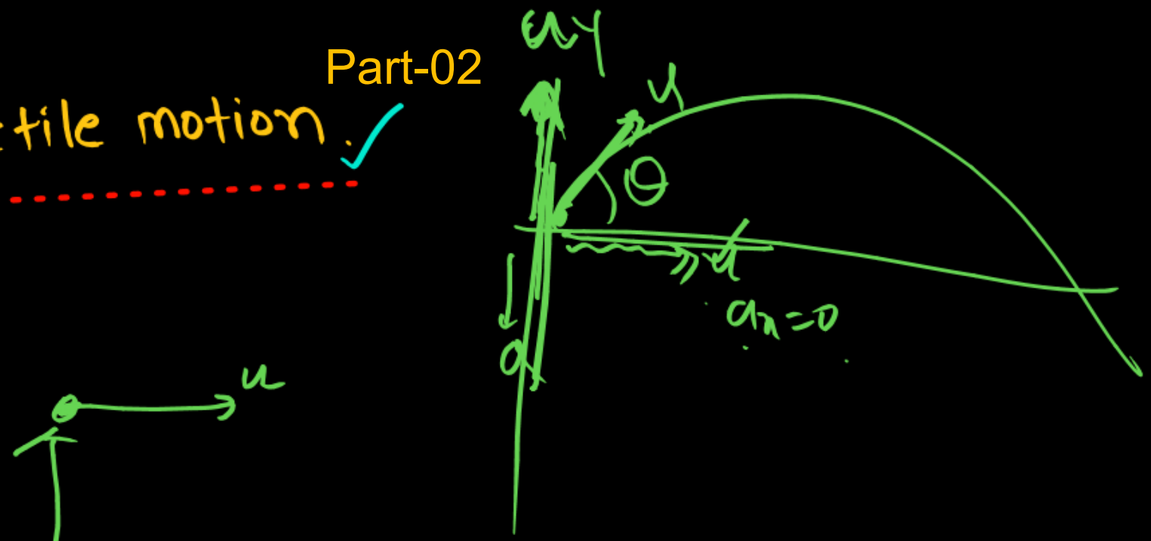
By- Manish Raj (MR Sir)

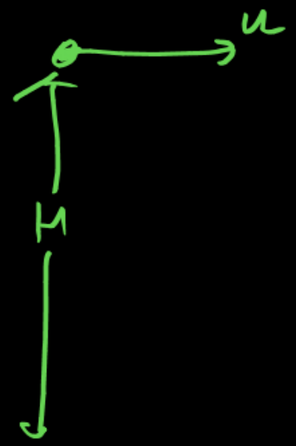


Todays Goal

Part-02

Horizontal Projectile motion





(2) object given a velocity $\vec{u}_{t=0} = 30\,m/s\,\hat{i}$ & accn of object $a = +5\,m/s^2\,\hat{j}$ ✓ find velocity of object at 2 sec.

initial

$u_x = 30\hat{i}$

$a_x = 0$

$V_x = u_x + a_x t$ ($a_x t^0$)

$V_x = 30\,\hat{i}$

$S_x = u_x t + \frac{1}{2} a_x t^2$ ($a_x t^0$)

$S_x = 30xt$

$u_y = 0$ ✓

$a_y = 5\,\hat{j}$ ✓

$V_y = u_y + a_y t$

$V_y = 0 + 5xt$

$\vec{V}_y = 5t\,\hat{j}$

$S_y = u_y t^0 + \frac{1}{2} a_y t^2$

$S_y = \frac{1}{2} 5t^2$

$\vec{S} = S_x i + S_y \hat{j}$

$= 30t\,i + \frac{5}{2} t^2 \hat{j}$

$\vec{V} = V_x i + V_y \hat{j}$

$\vec{V} = 30i + 5t\,\hat{j}$

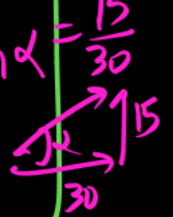
$V_{t=0} = 30i + 0\hat{j}$

$\vec{V}_{t=1} = 30i + 5\hat{j}$

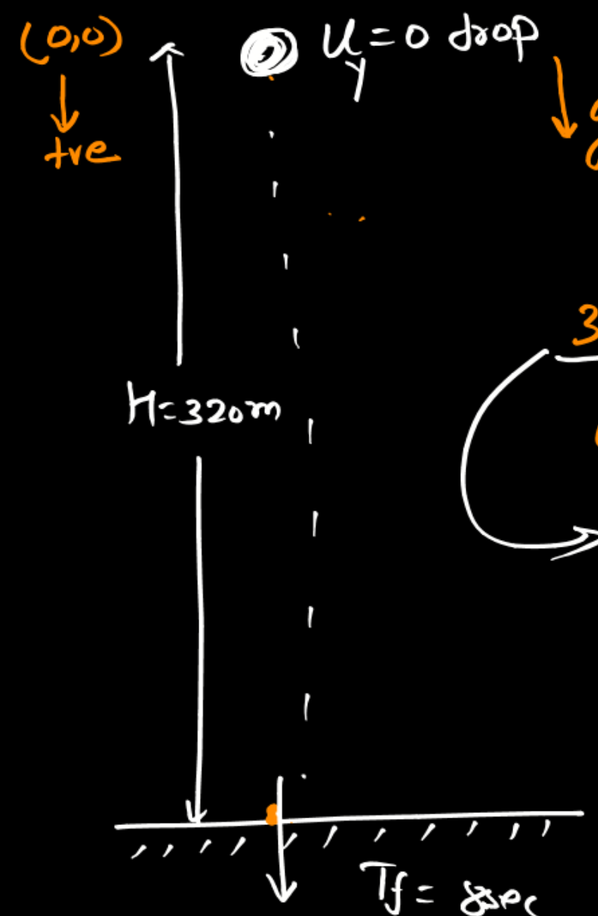
$V_{t=2} = 30i + 10\hat{j}$

$V_{t=3} = 30i + 15\hat{j}$ → $\tan\alpha = \frac{15}{30}$

$V_{t=4} = 30i + 20\hat{j}$



case-1



$$S = ut + \frac{1}{2}at^2$$

$$320 = 0 + \frac{1}{2}(10)t^2$$

$$64 = t^2$$

$$t = \sqrt{64} = 8\text{sec}$$

$$T = \sqrt{\frac{2H}{g}}$$

Velocity at ground

3rd eqn mo

$$v^2 - u^2 = 2as$$

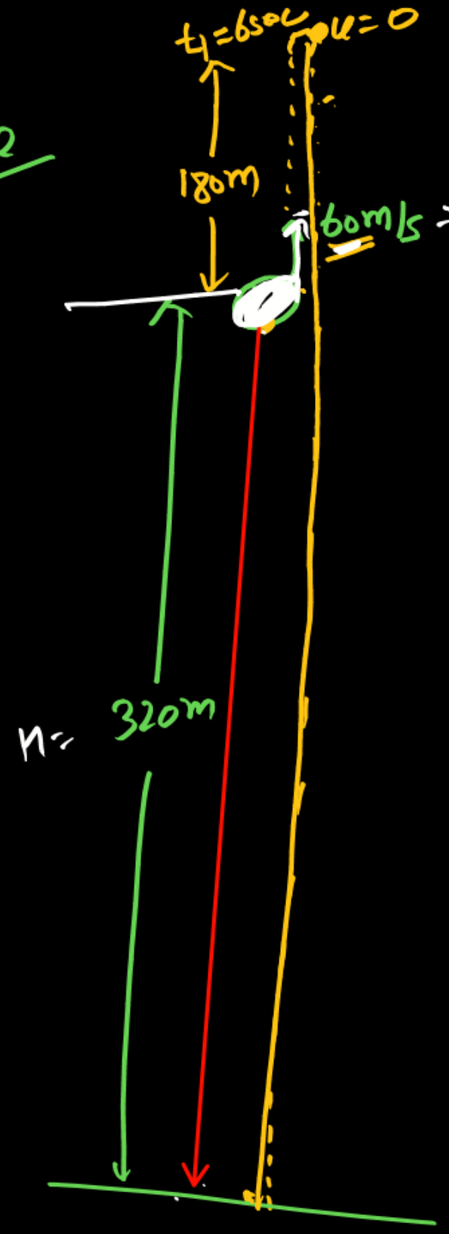
$$v = \sqrt{2 \times 10 \times 320}$$

$$= \sqrt{8400} = 80\text{ m/s}$$

$$V = u + at$$

$$= 0 + 10 \times 8 = 80\text{ m/s}$$

case-2



मध्यकिरण

$$S = ut + \frac{1}{2}at^2$$

$$+320 = -60t + \frac{1}{2}gt^2$$

$$5t^2 - 60t - 320 = 0$$

$$t^2 - 12t - 64 = 0$$

$$t^2 - 16t + 4t - 64 = 0$$

$$t(t-16) + 4(t-16) = 0$$

$t_1 = +16$ ✓

$t_2 = -4s$ ✗

MR*

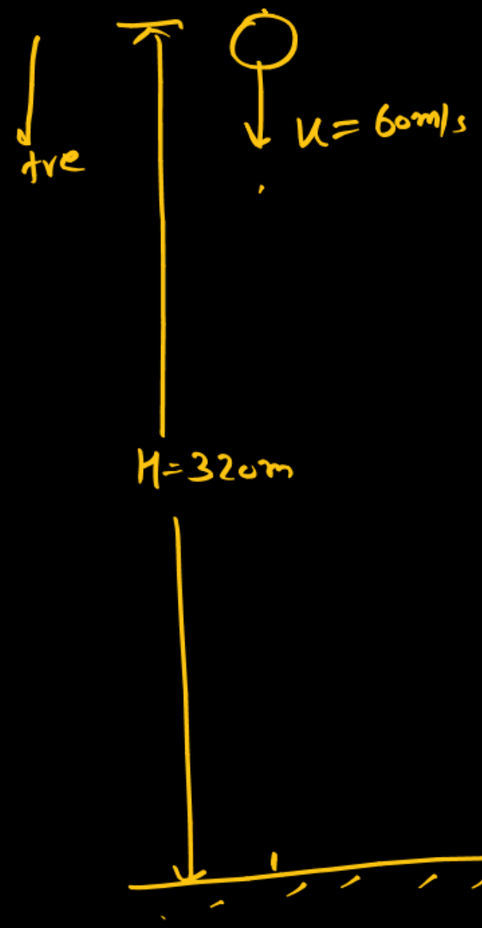
$t_1 = 6\text{sec}$

$$t_2 = \sqrt{\frac{2 \times 500}{10}} = \sqrt{100} = 10$$

$$T = t_1 + t_2 = 6 + 10$$

$$= 16\text{sec}$$

Case-3

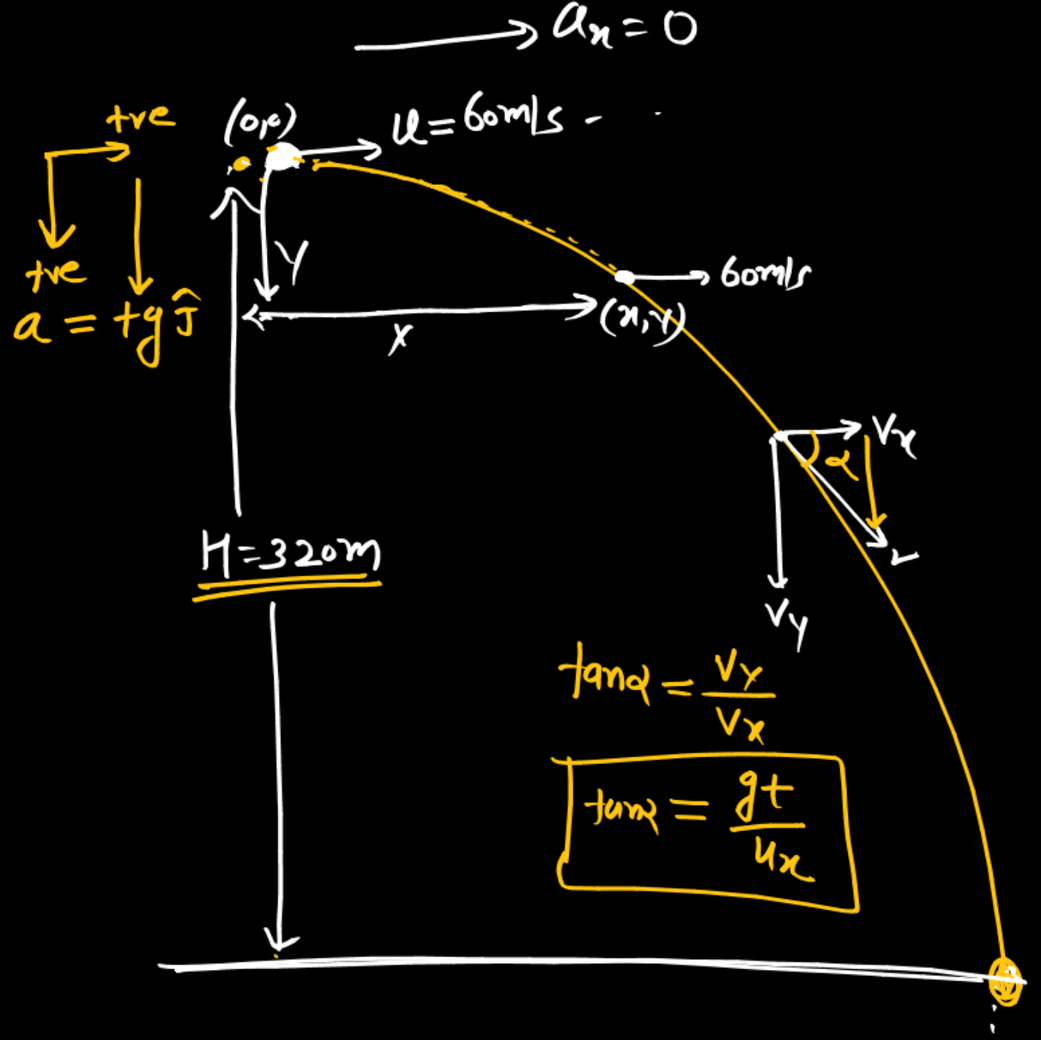


$$S = ut + \frac{1}{2}at^2$$

$$320 = 60t + \frac{1}{2}(10)t^2$$

$$5t^2 + 60t - 320 = 0$$

Horizontal Projectile motion



x-axis

$u_x = 60\,\hat{i}$

$a_x = 0$

$V_x = u_x = 60\,m/s\,\hat{i}$

$x_t = u_x t$

$x = 60t$ — (i)

y-axis

$u_y = 0$

$a_y = +g\,\hat{j}$

$= 10\,m/s^2\,\hat{j}$

$\vec{V}_{ty} = 10t\,\hat{j}$

$y = \frac{1}{2}\,10t^2$

$y = 5t^2$ — (ii)

$\vec{V} = 60\hat{i} + 10t\hat{j}$

$\vec{r} = x\hat{i} + y\hat{j}$

$= 60t\,\hat{i} + 5t^2\hat{j}$

eqn of Trajectory

$y = 5(t)^2$

Put t from (i)

$y = 5\left(\frac{x}{60}\right)^2$

↳ Parabola

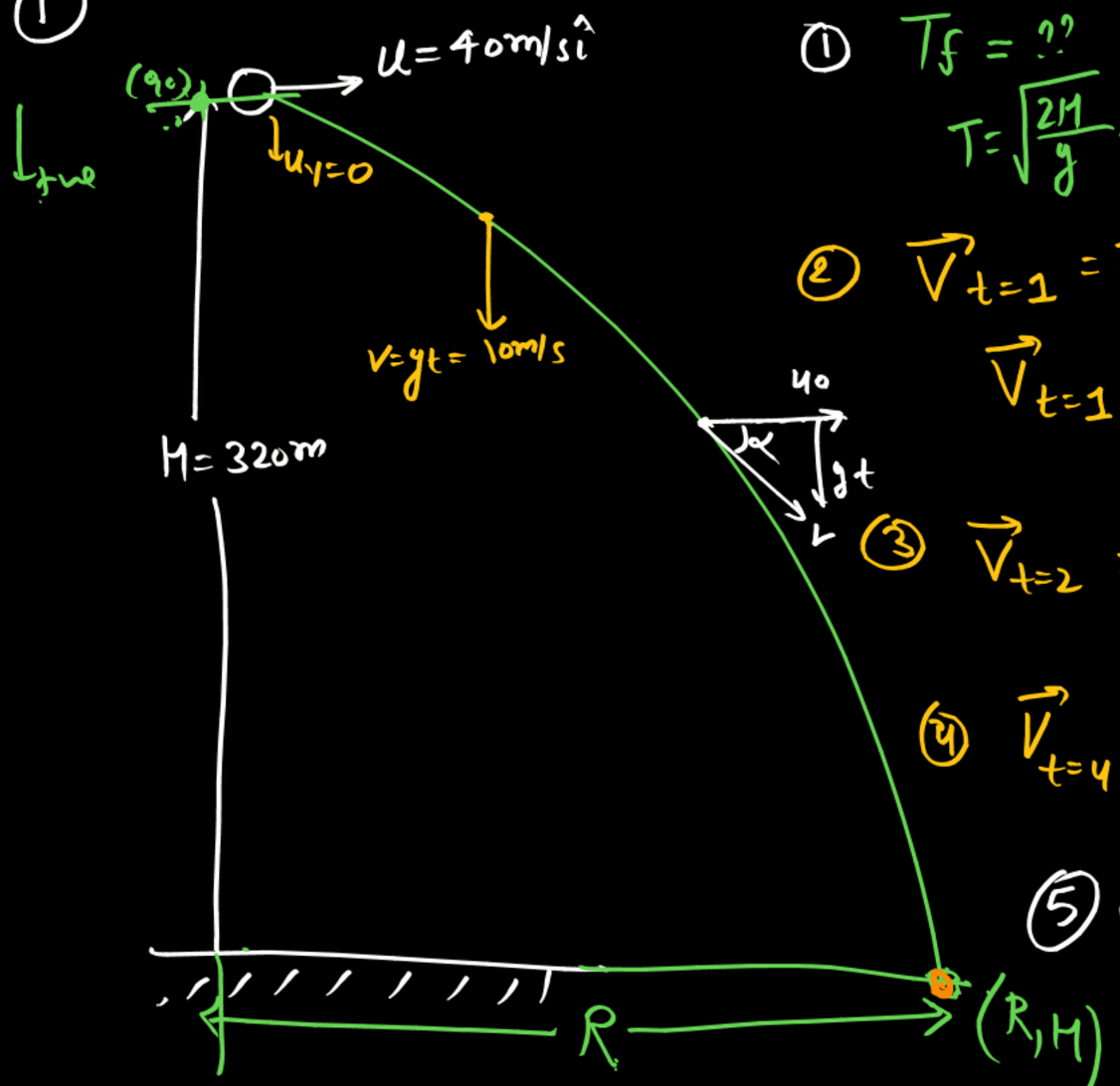
Cosinst motion in y-axis

$[S = ut + \frac{1}{2}at^2]$

$320 = 0 + \frac{1}{2}10\,t^2$

$t = \sqrt{64} = 8\,sec$ ✓

①



① $T_f = ??$

$T = \sqrt{\frac{2H}{g}} = \sqrt{\frac{2\times320}{10}} = \sqrt{64} = 8\,sec$

② $\vec{V}_{t=1} = ??$

$\vec{V}_{t=1} = 40\hat{i} + 10\hat{j}$

③ $\vec{V}_{t=2} = 40\hat{i} + 20\hat{j}$

④ $\vec{V}_{t=4} = 40\hat{i} + 40\hat{j}$

⑤ dirⁿ of velocity at 3sec from Horizon.

$\tan\alpha = \frac{gt}{40} = \frac{30}{40}$

$\alpha = 37°$

⑥ dispⁿ in t sec

$x = u_x t = 40t$

$y = \frac{1}{2} a_y t^2 = 5t^2$

⑦ eqⁿ of trajectory

$y = 5t^2 = 5\left(\frac{x}{40}\right)^2$ Parabola

⑧ dispⁿ in 4 sec

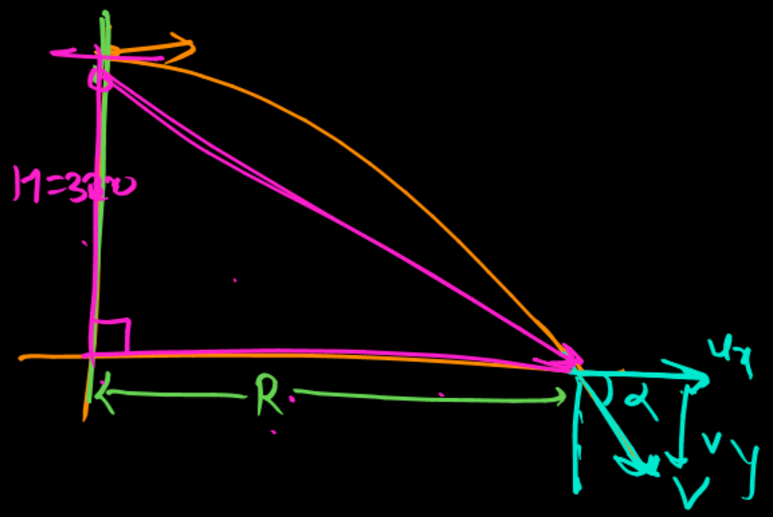
$x = 40\times4 = 160m\,\hat{i}$

$y = \frac{1}{2}10(4)^2\hat{j}$

⑨ Velocity at ground.

$\vec{V}_{gn} = 40\hat{i} + 80\hat{j}$ ✓

(10) Angle at Ball collide at ground.



$$\tan\alpha = \frac{v_y}{v_x}$$

$$\tan\alpha = \frac{80}{40}$$

$$\tan\alpha = 2$$

$$\alpha = \tan^{-1}(2)$$

(11) Range

$$R = u_x T_f$$

$$= 40 \times 8$$

$$= 320\ m$$

(12) Avg velocity in compt Jour

$$\vec{V}_{avg} = \frac{\vec{u}_i + \vec{v}_f}{2}$$

$$\vec{V}_{\Delta} = \frac{\text{Total displ}^n}{\text{total time}}$$

$$= \frac{320\sqrt{2}}{8}$$

$$= 40\sqrt{2}$$

$$= \frac{40\hat{i} + 40\hat{i} + 80\hat{j}}{2}$$

$$= \frac{80\hat{i} + 80\hat{j}}{2}$$

$$= 40\hat{i} + 40\hat{j}$$

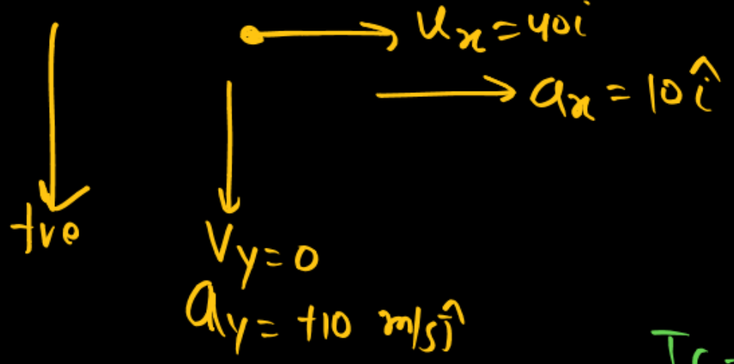
$$|V|_{Avg} = 40\sqrt{2}$$

(13) Chage in velocity

$$\Delta\vec{v} = \vec{v}_f - \vec{v}_i$$

$$= 80\hat{j}$$

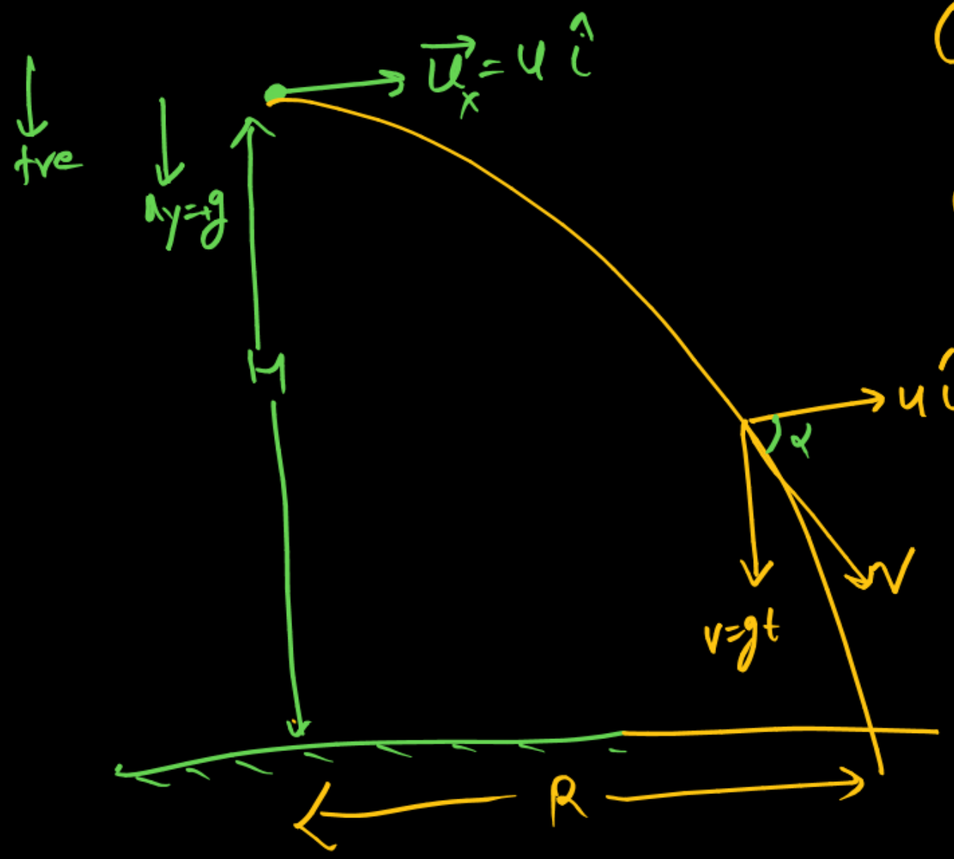
(14) If extra acc^n due to air $\vec{a}_x = 10\hat{i}$ is given then find R, T_f



$$T_f = \sqrt{\frac{2H}{g}} = 8\sec \checkmark$$

$$R = u_x T_f + \frac{1}{2} a_x t^2 = \left(40 \times 8 + \frac{1}{2} 10 (8)^2\right) m.$$

Horizontal Projection



① $T = \sqrt{\frac{2H}{g}}$

② $\vec{V}_t = u\hat{i} + gt\hat{j}$

③ speed $|\vec{V}| = \sqrt{u^2 + (gt)^2}$

④ dirn of motn at t

$\tan\alpha = \frac{V_y}{V_x}$

$\boxed{\tan\alpha = \frac{gt}{u}}$

⑤ $R_{any} = u \times T_f$

⑥ velocity at grnd

$\vec{V} = u\hat{i} + gT_f\hat{j}$

⑦ $x_t = ut \quad y = \frac{1}{2}gt^2$

⑧ eqn of Traj

$y = \frac{1}{2}g\left(\frac{x}{u}\right)^2$

$y = \frac{gx^2}{2u^2}$

$\boxed{y \propto x^2}$ Parabola

⑥



① $T_f = \sqrt{\frac{2 \times 180}{10}} = \sqrt{36} = 6 sec$

② $\vec{V}_t = 40i + gt\hat{j}$

③ $\vec{V}_{t=3} = 40i + 30\hat{j}$

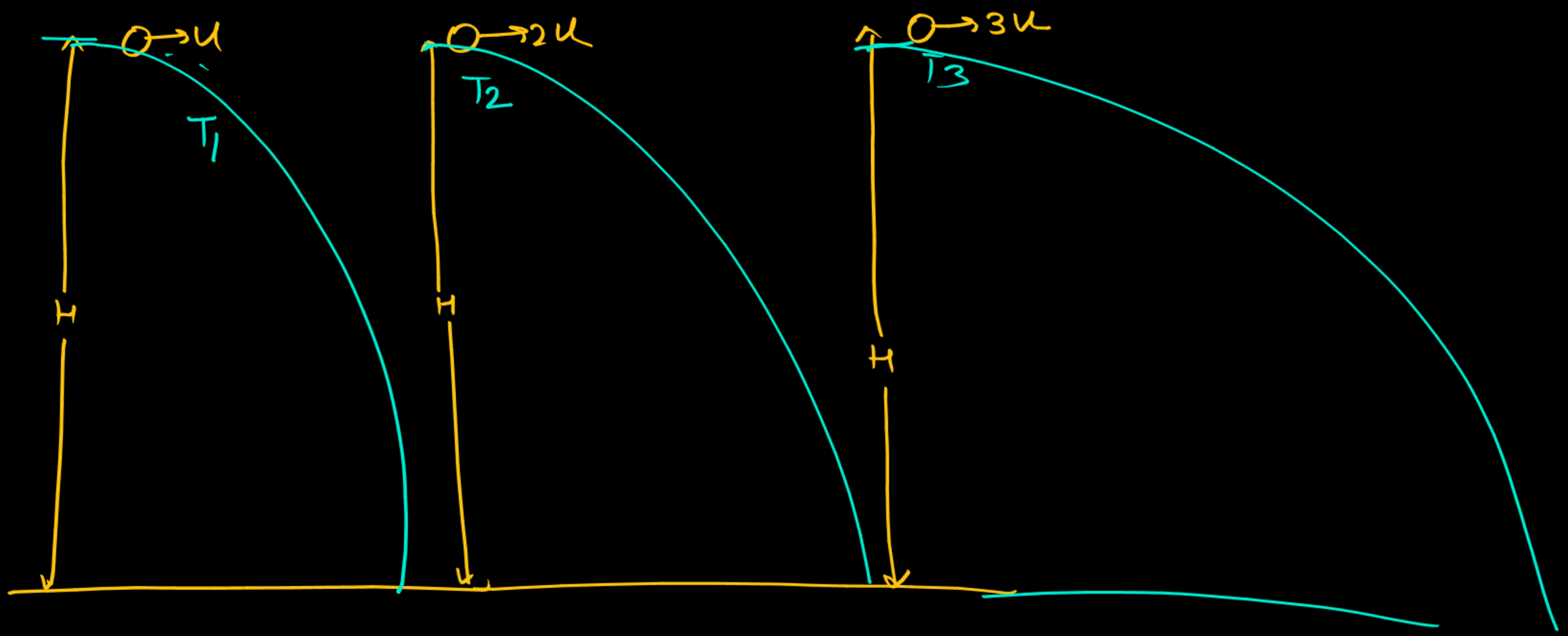
④ $R_y = 40 \times 6 = 240m$

$\vec{V}_{final} = 40i + 60\hat{j}$

Angle of collision from ground

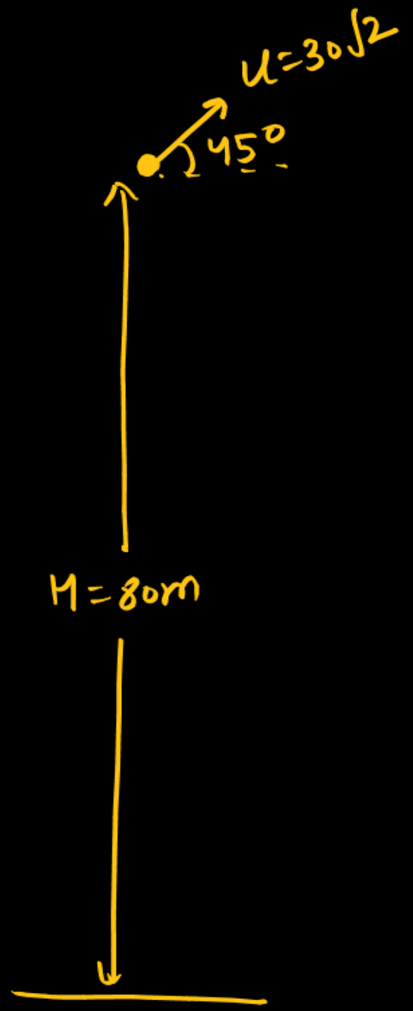
$\tan\alpha = \frac{gt}{u}$ ✓





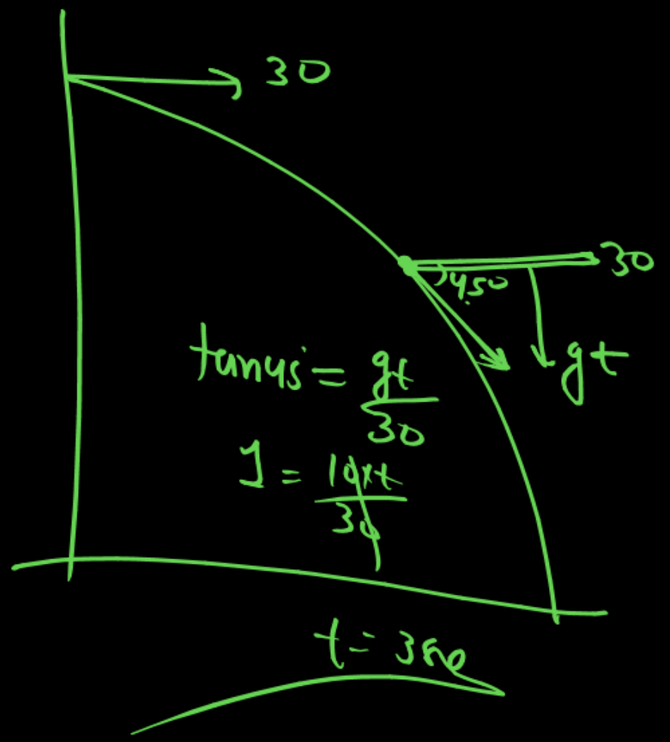
$\rightarrow$ $T_1 = T_2 = T_3$

$R_1 < R_2 < R_3$


u=30√2
45°
H=80m

Question 32

Ball is projected with 30 m/s in horizontal direction from some height. Find time when it is 45° from horizontal.



Question

When a particle is thrown horizontally, with initial velocity 'u' the resultant velocity of the projectile at any time t is given by:

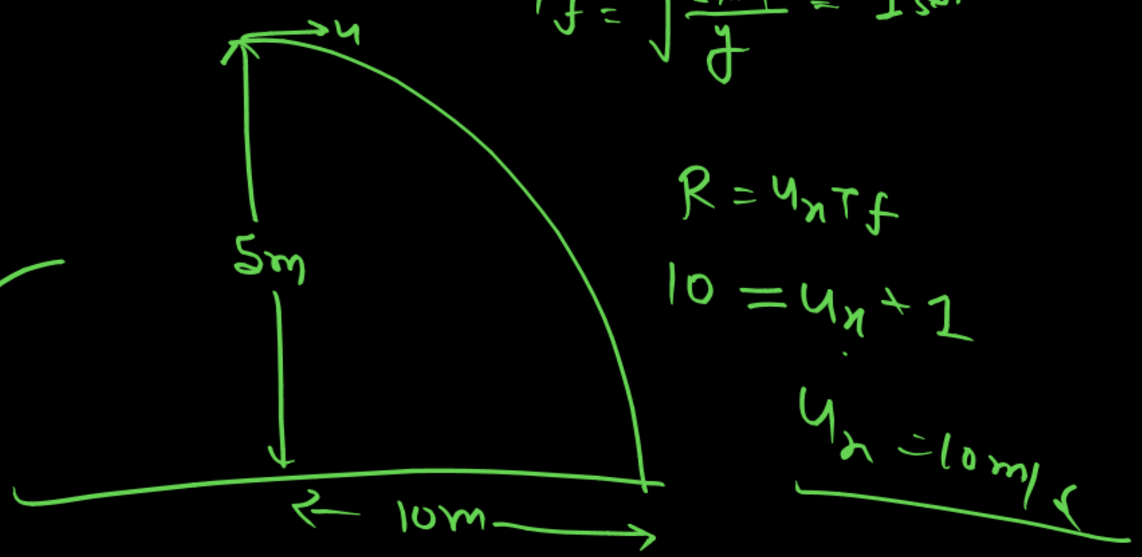
1. gt
2. $\frac{1}{2}gt^2$
3. $\sqrt{u^2 + g^2t^2}$
4. $\sqrt{u^2 - g^2t^2}$

$\vec{v} = ui + gt\hat{j}$

Question

A body is thrown horizontally from the top of a tower of height 5 m. It touches the ground at a distance of 10 m from the foot of the tower. The initial velocity of the body is (g = 10 ms)

1. 2.5 ms
2. 5 ms
3. 10 ms
4. 20 ms



$T_f = \sqrt{\frac{2\times 5}{g}} = 1$ sec.

$R = u_x T_f$

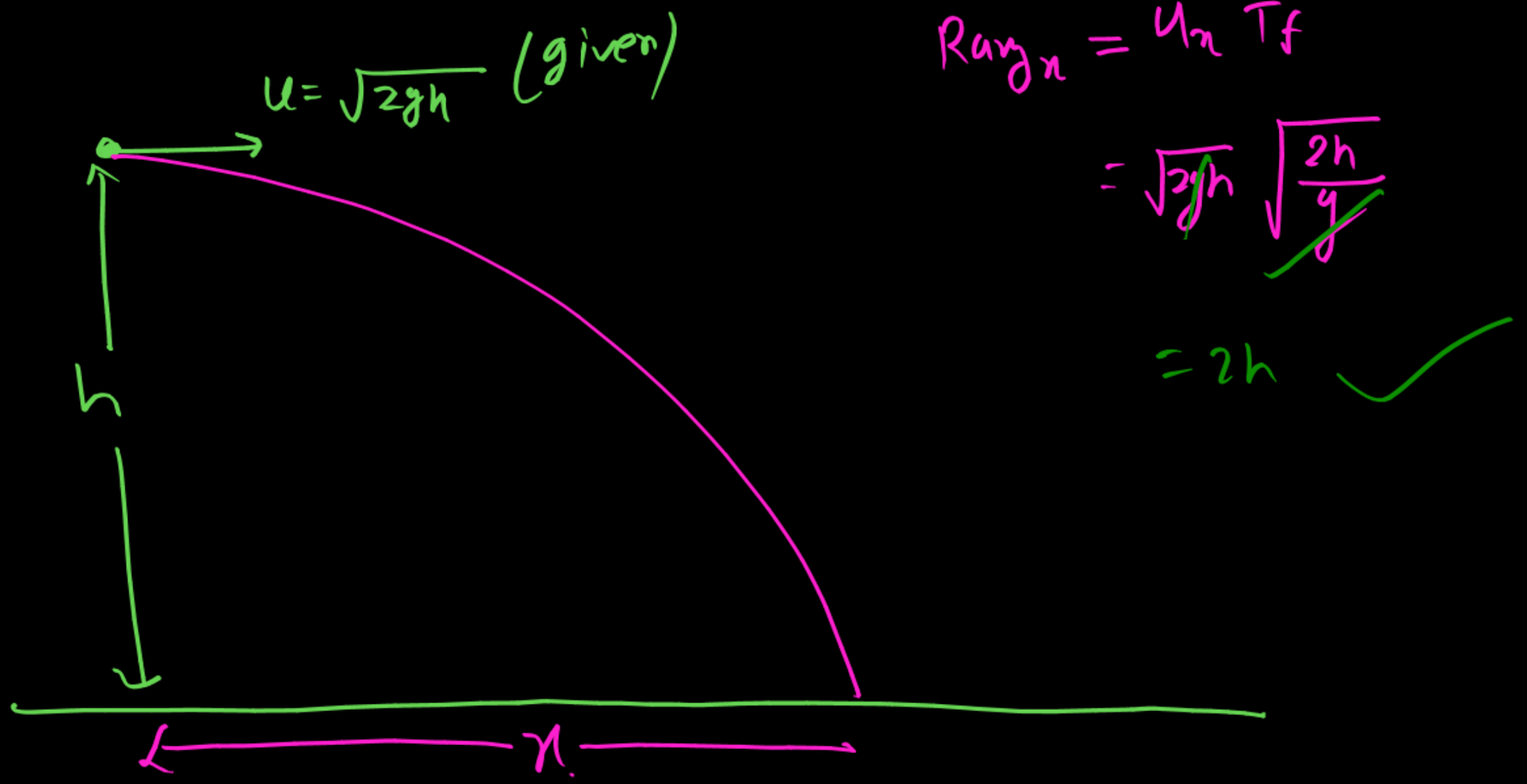
$10 = u_x \times 1$

$u_x = 10$ m/s

Question

A body is thrown horizontally with a velocity $\sqrt{2gh}$ from the top of a tower of height h. It strikes the level ground through the foot of the tower at a distance x from the tower. The value of x is:

1. h
2. $h/2$
3. $2h$
4. $2h/3$



$Range_x = u_x T_f$

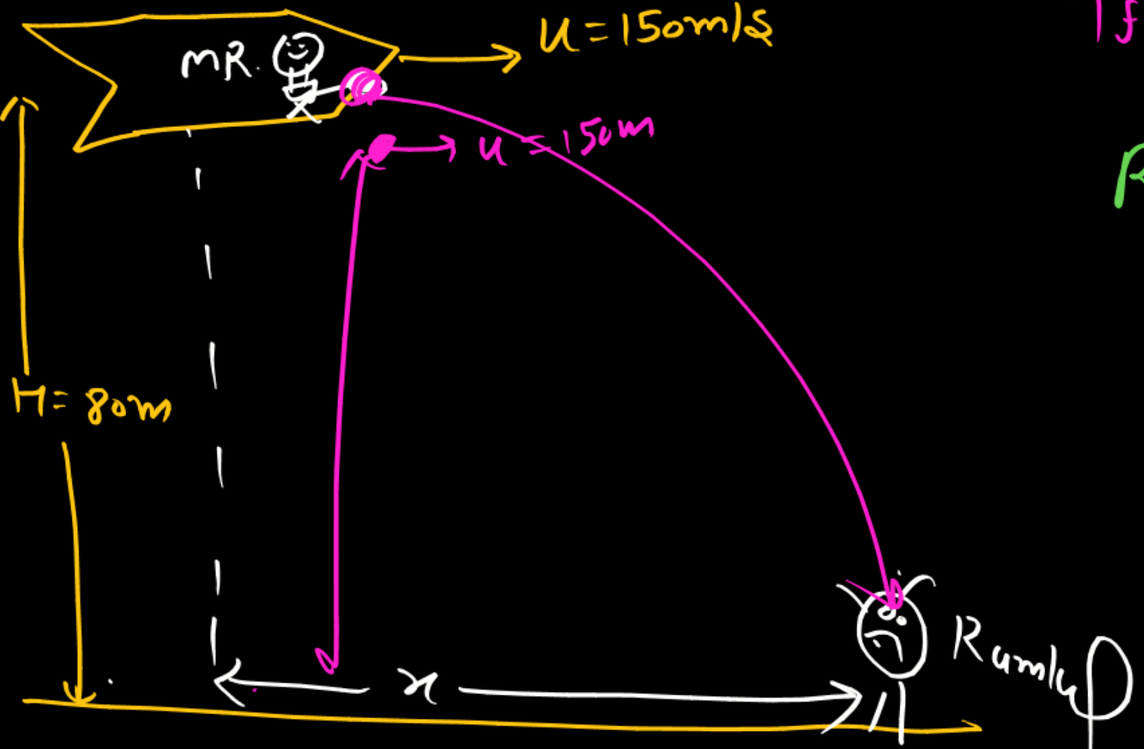
$= \sqrt{2gh}\sqrt{\frac{2h}{g}}$

$= 2h$

Question

A bomber is flying horizontally with a constant speed of 150 m/s at a height of 78.4 m. The pilot has to drop a bomb at the enemy target. At what horizontal distance from the target should he release the bomb:

1. Zero
2. 300 m
3. 600 m
4. 750 m



$T_f = \sqrt{\frac{2H}{g}} = 4\,\text{sec}$

$R = u_x T_f$

$= 150 \times 4 = 600\,\text{m}$

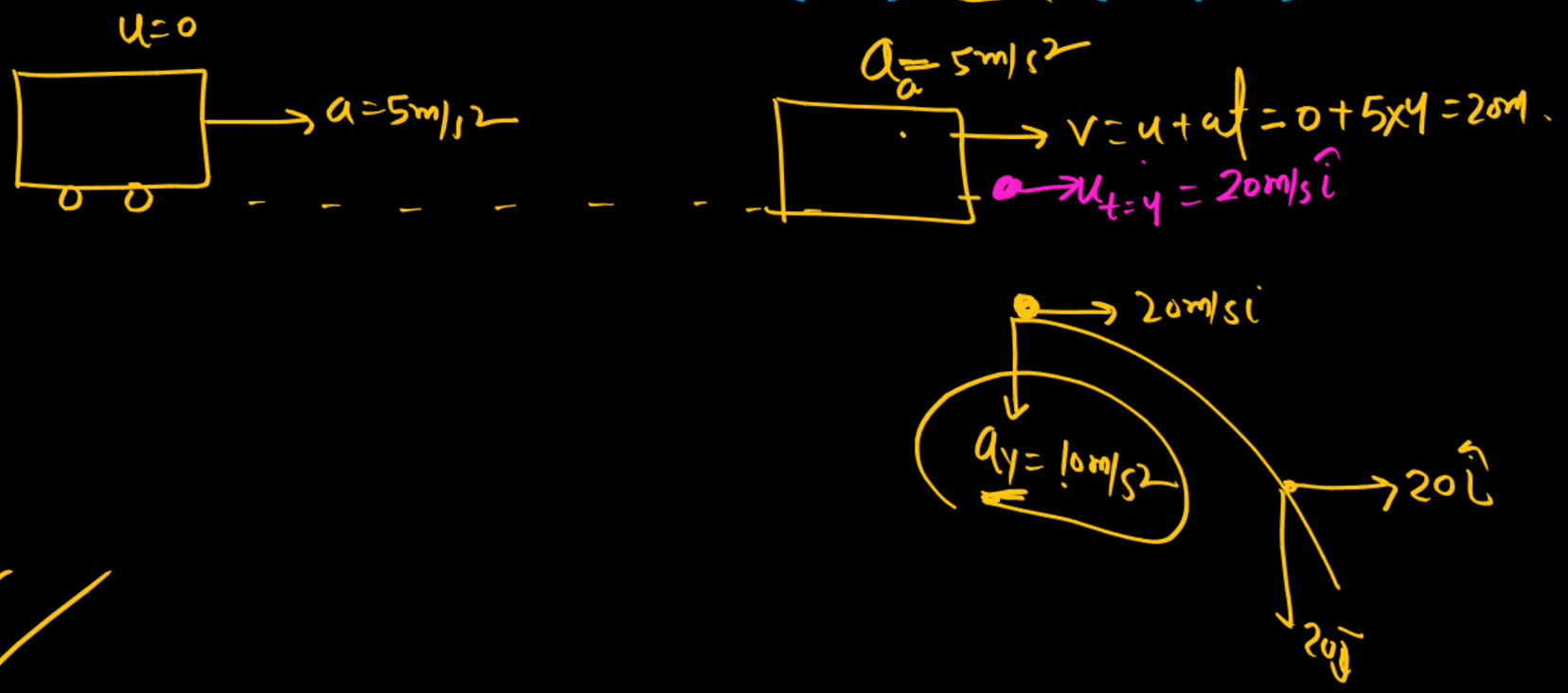
Question

A car starts from rest and accelerates at 5 m/s^2. At $t = 4$ s, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at $= 6$ s ?

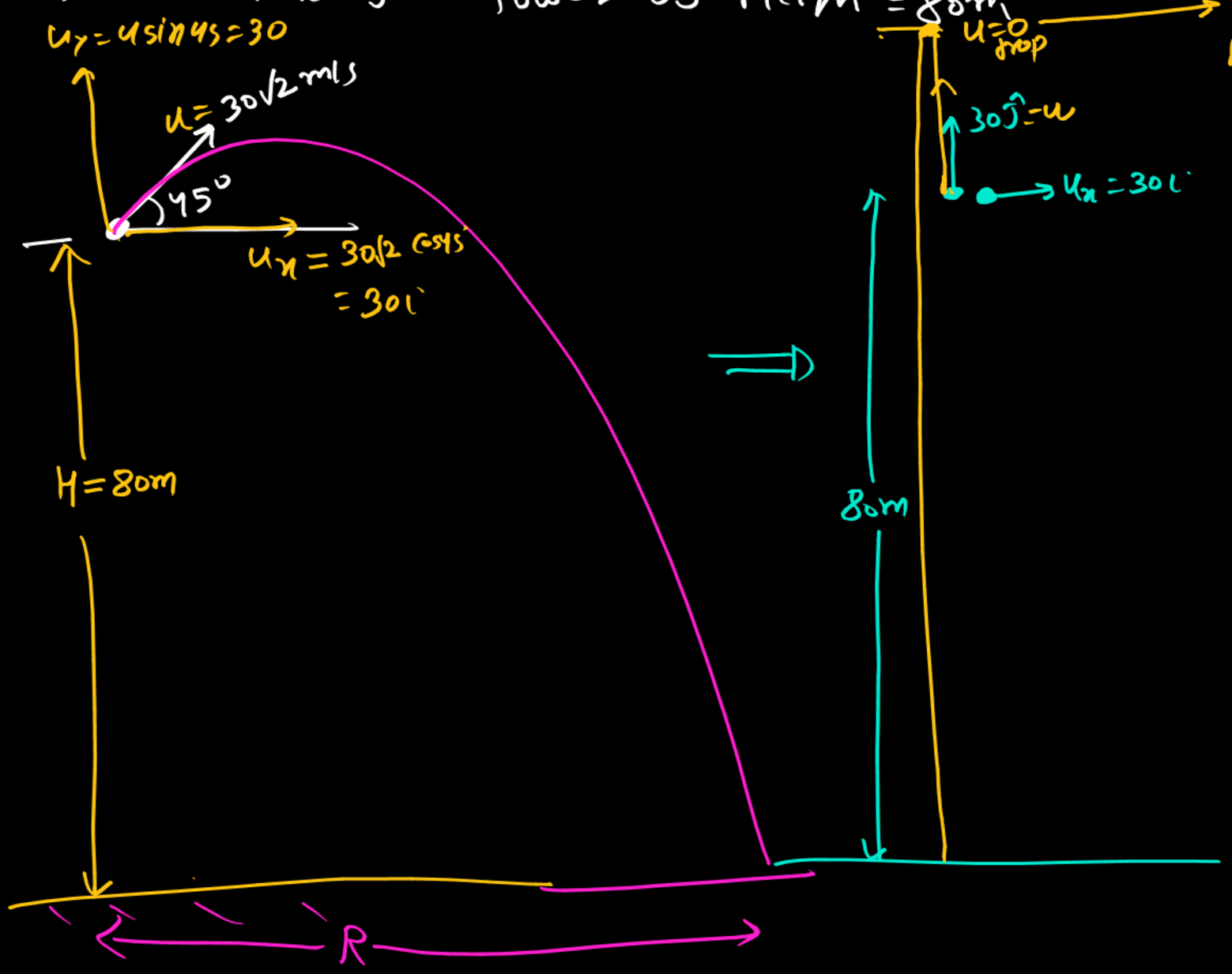
(Take g $= 10\ m/s^2$)

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1. 20 m/s, 5 m/s^2
2. 20 m/s, 0
3. $20\sqrt{2}$ m/s, 0
4. $20\sqrt{2}$ m/s, 10 m/s^2



Ball is Projected at 45° from Tower of Height = 80m



$t_1 = 3$sec

$h_{max} = 45$m

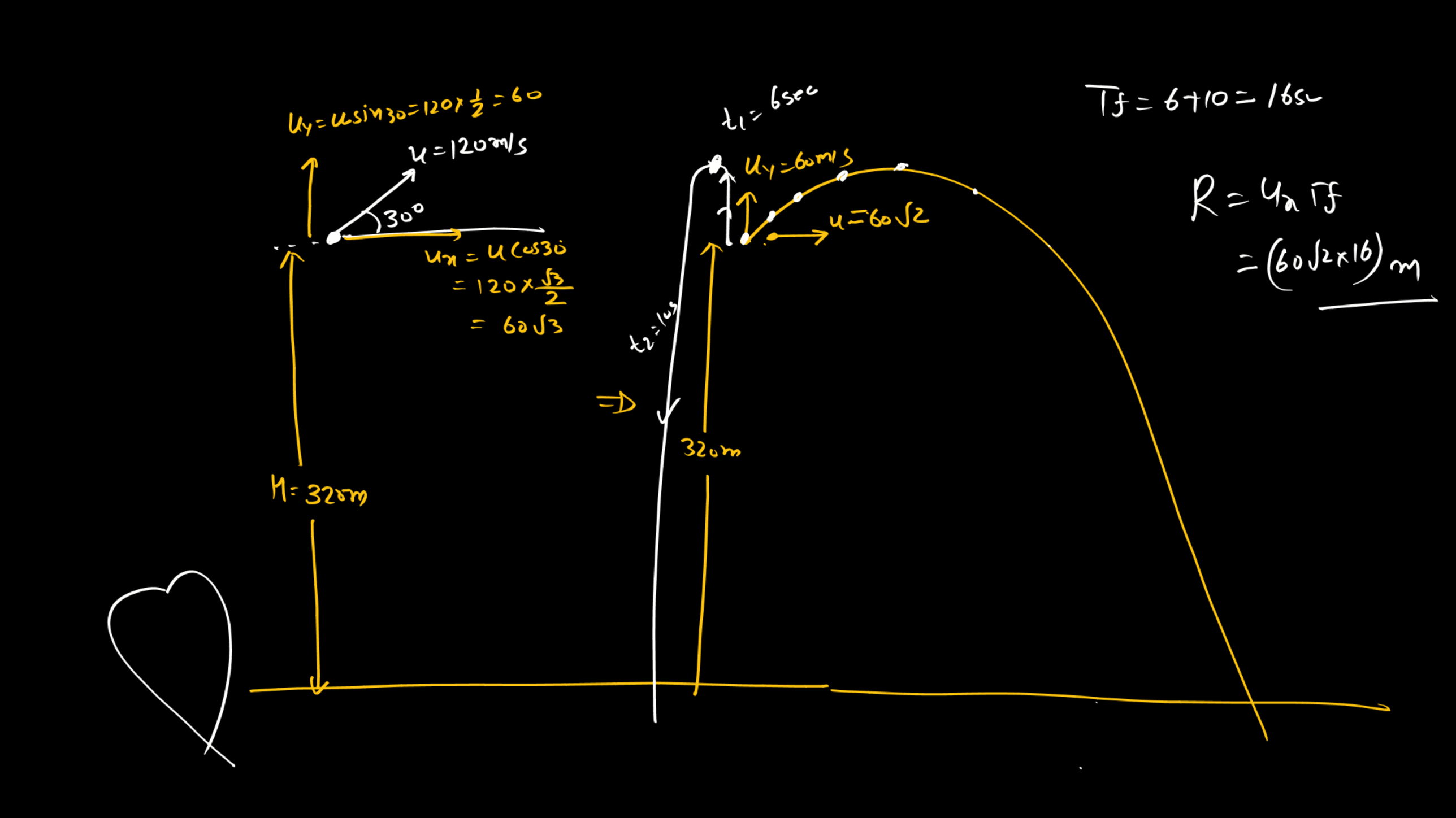
$T_2 = 5$sec

$T_f = 3 + 5 = 8$sec ✓

$R_{ng} = u_x T_f$

$= 30 \times 8$

$= 240$m


$u_y = u\sin 30 = 120 \times \frac{1}{2} = 60$
$u = 120\ m/s$
30°
$u_x = u\cos 30$
$= 120 \times \frac{\sqrt{3}}{2}$
$= 60\sqrt{3}$
$H = 320m$
$t_1 = 6sec$
$u_y = 60\ m/s$
$u = 60\sqrt{2}$
$t_2 = 10s$
$320m$
$T_f = 6 + 10 = 16s$
$R = u_x T_f$
$= (60\sqrt{2} \times 16)\ m$

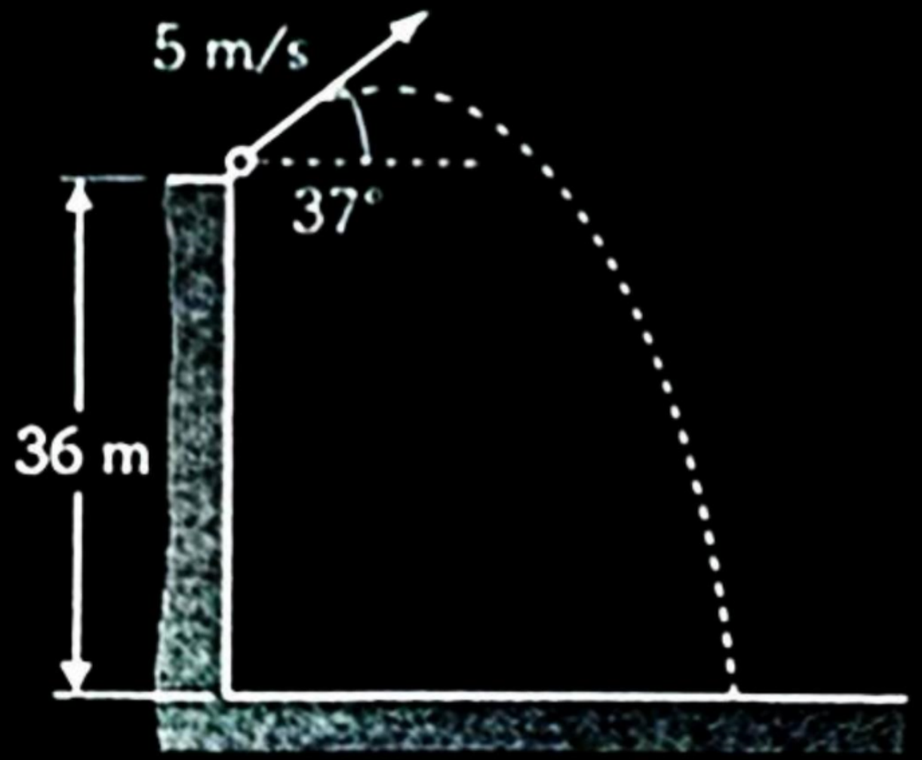
Question

A ball is projected upwards from the top of a tower with a velocity of 50 m/s making an angle of 30° with the horizontal. The height of the tower is 70 m. After how much time from the instant of throwing, will the ball reach the ground?

1. 2 s
2. 5 s
3. 7 s
4. 9 s

Question

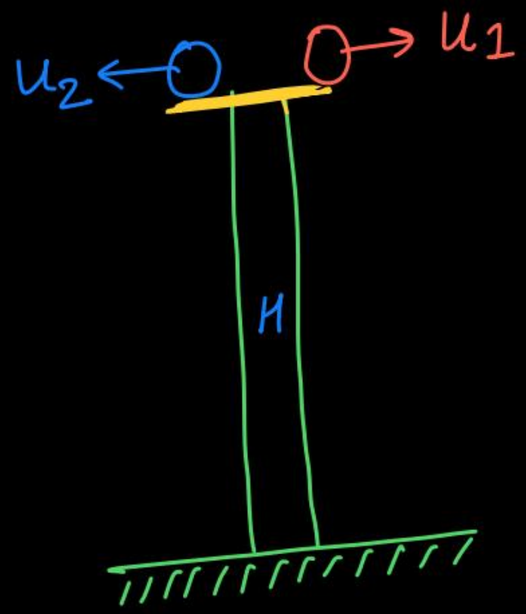
A ball is thrown from the top of 36 m high tower with velocity 5 m/s at an angle 37° above the horizontal as shown. Its horizontal distance on the ground is closest to [$g = 10\ m/s^2$]



1. 12 m
2. 18 m
3. 24 m
4. 30 m

Two Ball Projected Horizontaly from Height H. in opposite direction with speed u_1 and u_2 then find time when they moving perpendicular to each other and also find Horizontal distance between them when they are moving perpendicular

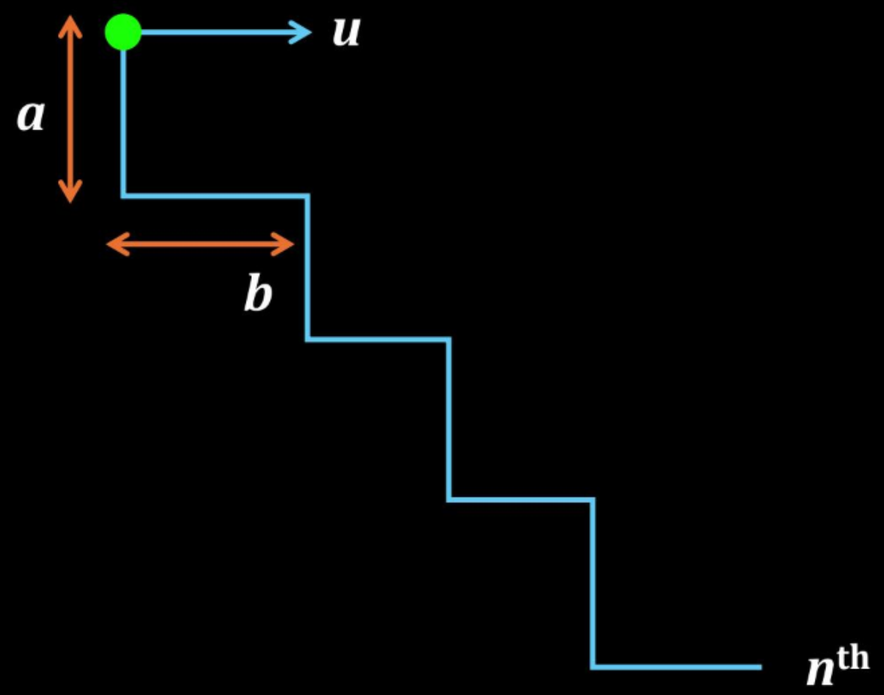
(3) जमप्
mRV



Question

Find velocity so that ball will fall on n^{th} step.

(4)



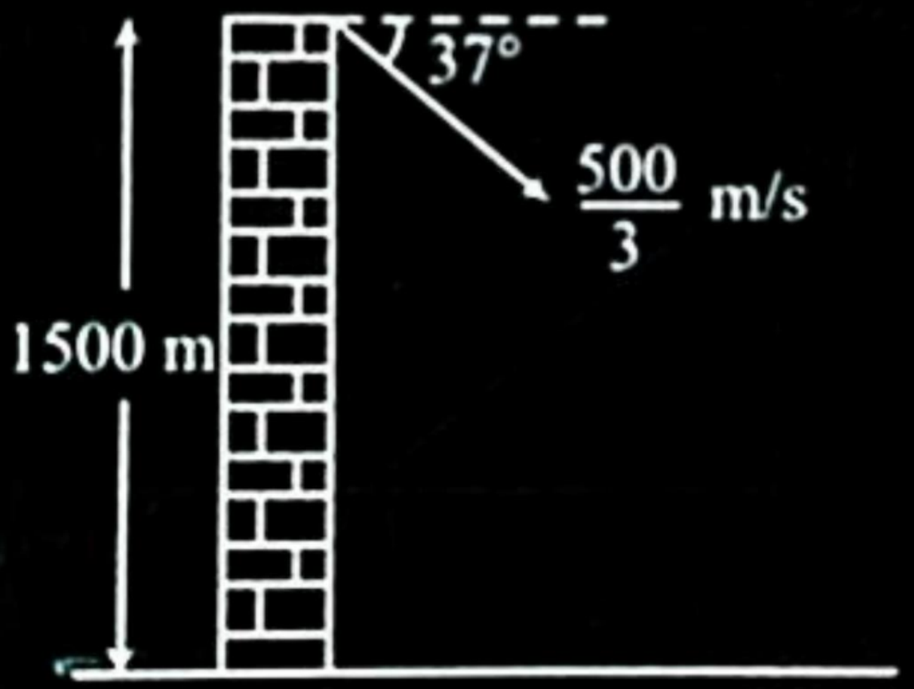
Question

8

A particle is projected from a tower as shown in figure, then the distance from the foot of the tower where it will strike the ground will be:

(take $g = 10\ m/s^2$)

1. 4000/3 m
2. 5000/3 m
3. 2000 m
4. 3000 m



Thank You